

LUEDENSCHEID, Germany

WMO: 104180

Lat: 51.2453N

Lon: 7.6425E

Elev: 389

StdP: 96.73

Time Zone: 1.00 (EUC)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.3	-7.2	-13.2	1.3	-6.2	-10.8	1.6	-5.4	11.0	5.3	9.9	5.8	2.3	80	0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.3	27.7	18.1	25.7	17.5	23.9	16.9	19.5	25.1	18.5	23.7	17.7	22.3	2.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.6	13.2	21.0	16.7	12.5	20.1	15.9	11.8	19.5	57.0	25.5	54.0	23.7	51.2	22.5	22.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.5	7.4	6.5	DB	-12.3	31.2	3.4	1.5	-14.8	32.3	-16.7	33.2	-18.6	34.1	-21.1	35.2	(4)
(5)			WB	-12.7	21.2	3.4	0.8	-15.1	21.8	-17.0	22.3	-18.9	22.8	-21.4	23.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.7	0.9	1.5	4.2	8.1	12.1	14.6	16.7	16.5	13.2	9.4	5.0	1.4	(6)	
	DBStd	6.88	4.29	4.23	4.05	4.16	3.94	3.65	3.44	3.24	3.25	3.70	3.57	4.09	(7)	
	HDD10.0	1301	284	237	184	85	24	3	0	0	6	57	154	267	(8)	
	HDD18.3	3589	542	471	439	307	197	122	76	76	157	277	401	526	(9)	
	CDD10.0	813	0	1	3	28	88	141	207	203	103	38	3	0	(10)	
	CDD18.3	60	0	0	0	0	3	9	24	20	4	0	0	0	(11)	
	CDH23.3	509	0	0	0	2	24	92	193	175	23	1	0	0	(12)	
	CDH26.7	104	0	0	0	0	2	16	43	42	1	0	0	0	(13)	
(14) Wind	WSAvg	2.9	3.3	3.5	3.2	2.7	2.7	2.6	2.6	2.4	2.5	2.9	2.9	3.2	(14)	
(15) Precipitation	PrecAvg	1223	130	104	93	68	79	91	107	106	94	96	117	137	(15)	
	PrecMax	1632	259	270	205	120	179	164	184	235	220	311	204	317	(16)	
	PrecMin	917	12	29	12	4	30	20	44	37	23	38	5	38	(17)	
	PrecStd	181	64	58	53	29	37	32	44	54	47	53	51	65	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.7	12.4	16.9	23.2	26.0	28.9	30.2	30.5	26.0	20.6	14.4	11.0	(19)	
		MCWB	9.0	7.9	10.1	13.1	16.2	18.5	18.9	19.8	17.2	15.5	10.7	9.3	(20)	
	2%	DB	9.3	10.0	14.4	20.3	23.2	25.9	27.7	27.4	23.4	18.1	12.5	9.7	(21)	
		MCWB	8.0	7.3	8.9	12.3	15.2	17.6	18.1	17.8	17.0	14.1	10.2	8.5	(22)	
	5%	DB	8.0	8.6	12.2	17.7	21.3	23.7	25.4	25.2	20.8	16.1	11.1	8.5	(23)	
		MCWB	7.0	6.8	8.3	11.1	14.1	16.7	17.6	17.7	15.7	13.3	9.7	7.5	(24)	
	10%	DB	6.4	7.3	10.1	15.4	19.3	21.5	23.3	22.9	18.7	14.6	9.8	7.0	(25)	
		MCWB	5.4	6.0	7.5	10.1	13.3	15.8	17.0	17.0	14.7	12.5	8.5	6.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.8	9.2	11.5	14.2	18.2	20.1	20.5	20.6	19.0	16.3	11.7	10.1	(27)	
		MCDB	10.1	10.4	14.8	20.4	23.3	26.1	27.0	28.0	23.1	19.5	13.0	10.7	(28)	
	2%	WB	8.2	8.0	10.0	13.0	16.6	18.7	19.4	19.3	17.5	14.7	10.6	8.8	(29)	
		MCDB	8.8	9.4	13.0	18.7	21.6	23.9	24.8	25.3	22.0	17.0	11.9	9.5	(30)	
	5%	WB	6.9	7.1	8.9	11.9	15.2	17.5	18.5	18.3	16.4	13.7	9.8	7.5	(31)	
		MCDB	7.8	8.2	11.2	16.5	19.6	22.0	23.5	23.2	20.0	15.7	11.0	8.3	(32)	
	10%	WB	5.5	6.0	7.7	10.7	13.8	16.4	17.5	17.3	15.1	12.6	8.5	6.2	(33)	
		MCDB	6.3	7.1	9.8	14.8	18.0	20.4	22.1	21.5	18.0	14.3	9.6	6.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.3	4.8	6.5	8.2	8.5	8.7	8.3	8.4	7.3	6.2	4.3	4.0	(35)	
		MCDBR	5.1	6.5	10.6	12.2	12.0	12.4	12.1	12.3	10.8	8.1	5.7	4.9	(36)	
	5% WB	MCWBR	4.4	4.5	6.1	6.2	6.0	5.8	5.0	5.4	5.5	4.9	4.2	4.2	(37)	
		MCWBR	4.9	5.0	8.3	10.4	10.5	10.7	10.8	10.4	9.5	7.0	5.0	4.4	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.307	0.331	0.361	0.394	0.399	0.400	0.408	0.402	0.373	0.348	0.328	0.304	(40)	
		taud	2.419	2.378	2.319	2.265	2.281	2.294	2.270	2.312	2.383	2.442	2.428	2.402	(41)	
	Ebn at Noon	Ebn at Noon	717	793	832	843	855	856	841	826	811	764	681	654	(42)	
		Edh at Noon	60	81	103	122	125	125	126	115	96	75	60	53	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.73	1.41	2.53	4.06	4.88	5.28	5.02	4.28	3.13	1.84	0.86	0.56	(44)	
	RadStd	RadStd	0.11	0.26	0.38	0.47	0.46	0.48	0.63	0.40	0.38	0.17	0.12	0.07	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (64 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page